OSINT INVESTIGATION REPORT

東北電子産業株式会社 (Tohoku Electronic Industrial Co., Ltd.)

IP, Chemiluminescence Technology and JPO:METI Recognition

Japan

Subject	東北電子産業株式会社 (Tohoku Electronic Industrial Co., Ltd.)
Key individual	山田理恵 (Rie Yamada), 代表取締役社長
Investigation period	2000–2026
Case type	IP / technology recognition
Source basis	Public Japanese sources

RESEARCH QUESTION

What specific technology/IP did 東北電子産業 develop, protect and commercialize, and how is it connected to the 2026 JPO/METI recognition of Rie Yamada?

Prepared from public sources only. No private, paid or non-public data was used.

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1. Executive Summary

The investigation identifies a specific technology at the center of東北電子産業株式会社's intellectual-property and standardization activity: high-sensitivity evaluation of oxidative degradation of materials through detection of chemically induced ultra-weak photon emission (UPE/chemiluminescence) . The technology is commercially embodied in the company's Chemiluminescence Analyzer (CLA) product family. The principle is based on detecting extremely weak light generated during oxidation reactions in organic materials. According to the company's technical explanation, oxidation produces peroxides; decomposition of these intermediates generates excited molecular species, including excited carbonyls and singlet oxygen, which emit light as they return toward the ground state. Measuring this emission provides an indirect but highly sensitive indicator of the material's oxidation state.ケミルミネッセンスアナライザー公式サイトThe technology is particularly relevant to polymers and plastics , including recycled plastics. The company's CLA systems are designed to detect oxidation at a very early stage, before conventional indicators such as significant mechanical-property degradation or visible discoloration necessarily become apparent. METI's Go-Tech documentation confirms development of a dedicated weak-light detection system for quantitative evaluation of oxidation degradation in recycled materials, including a high-temperature sample chamber, temperature-control functions and analytical software capable of calculating OIT, OOT and activation energy. Чуco The IP trail provides two particularly relevant Japanese patent publications. JP3813140B2 , originally filed by東北電子産業, concerns a method for measuring the degree of oxidation of a sample. Its description directly references the company's CLA-FS1 equipment. JP2002195951A , also filed by東北電子産業, concerns determining material deterioration through temperature-dependent chemiluminescence and lists Rie Yamada as a co-inventor . The latter also directly connects the methodology to CLA equipment. Патенты Google The technology subsequently moved beyond proprietary equipment into formal standardization. Through Japan's新市場創造型標準化制度(New Market Creation-type Standardization System),東北電子産業's technology became JIS K 7351 , concerning high-sensitivity measurement of weak emission from peroxides contained in plastics. METI records identify東北電子産業as the organization behind the 2016 standardization proposal, with JIS K 7351 publicly issued in January 2018. Минэкономторг Японии The international extension is ISO 4765:2022 , Chemically-induced ultra-weak photon emission (UPE) - Measurement as an analysis method of degradation of polymeric material . ISO describes the method as a means of assessing the very early oxidative degradation state of polymers . The standard was published in April 2022. ISO This sequence is directly relevant to the 2026 JPO recognition of山田理恵. Importantly, JPO did not award東北電子産業a corporate patent award . The 2026 award was a personal経済産業大臣表彰under the category知的財産権制度関係功労者and identifies山田理恵as president of東北電子産業. JPO's official material specifically records her statement identifying the weak-light material oxidation-evaluation technology as one of the company's significant achievements and noting that it was taken through the New Market Creation-type Standardization System to JIS and ISO standards. Японское агентство по патентам The evidence therefore supports the following model: Technical invention / IP → Chemiluminescence Analyzer → industrial oxidation/degradation analysis → JIS K 7351 → ISO 4765 → contribution to IP/standardization policy and practice → 2026 recognition of Rie Yamada The award should therefore be understood as recognition of Yamada's broader contribution to Japan's intellectual-property system and standardization , with the company's technology and its transition from proprietary technology to national and international standards serving as a concrete example-not as a prize granted solely for one patent.

2. Subject and Award Context

2.1. Identification of the recipient

The Japan Patent Office's official 2026 Chizai Koro-sho material identifies:山田理恵-東北電子産業株式会社代表取締役社長as a recipient of the経済産業大臣表彰in the category:知的財産権制度関係功労者(Persons who have contributed to the intellectual-property-rights system). Японское агентство по патентам This distinction is critical for interpreting the case. The same JPO page separately lists companies receiving corporate recognition under categories such as:

- 知財活用企業(特許)
- IP-utilizing company (patent);
- 知財活用企業(商標)
- IP-utilizing company (trademark);
- 知財活用ベンチャー
- IP-utilizing venture;
- other corporate categories.東北電子産業is not identified there as the recipient of a corporate patent or trademark award . The recipient relevant to this investigation is山田理恵personally. Японское

агентство по патентам Consequently, the investigation should not frame the case as: "東北電子産業won a JPO award for its patent." The more accurate formulation is: Rie Yamada, president of Tohoku Electronic Industrial, received the 2026 METI Minister's Award for contributions to the intellectual-property system; the company's chemiluminescence technology and its subsequent JIS/ISO standardization form an important concrete component of that record.

2.2. Why JPO recognized

山田理恵

The official JPO profile describes a broader record than the company's technology alone. According to JPO, Yamada participated over an extended period as a representative of regional SMEs in Japan's intellectual-property policy process, including the Cabinet Office's知的財産戦略本部. The JPO profile credits her with participation in policy formulation and discussion of important IP measures and with communicating the practical problems faced by regional SMEs. Японское агентство по патентам She also served on several IP-related committees, including:

- 検証・評価・企画委員会;
- 構想委員会;
- 地域価値ワーキンググループ. The JPO specifically notes that within the構想委員会she introduced her company's experience with JIS standardization and advocated accelerating development of human resources capable of advancing international standardization. Японское агентство по

патентам Thus, the award is structurally broader than a technology award. The technology becomes important because it demonstrates how Yamada's own company applied IP and standardization strategy in practice.

3. Technology Under Investigation

3.1. The technological object

The specific technology named by Yamada in the official JPO ceremony material is:微弱発光検出による材料の高感度酸化劣化評価装置A technically appropriate English rendering is: High-sensitivity apparatus for evaluating oxidative degradation of materials through ultra-weak photon-emission detection. This is the central technological object of the investigation. The commercial implementation is the company's Chemiluminescence Analyzer (CLA) .東北電子産業describes the CLA as an ultra-sensitive weak-light measurement system capable of detecting emissions at approximately 50 photons/cm2/sec , corresponding to approximately the watt level. The company presents this as a photon-level detection 10 −14 capability for light invisible to the human eye. Tohoku Electronic Industrial Co., Ltd.

3.2. Physical measurement principle

The technology does not literally "measure oxidation" as a chemical concentration sensor would. Instead, it measures light generated as a consequence of oxidation chemistry . The company's technical explanation describes the sequence approximately as follows: 1. An organic material reacts with oxygen. 2. Oxidation generates peroxide species, including hydroperoxides. 3. These species can decompose under heat or light. 4. The decomposition produces excited molecular species, including excited carbonyls and singlet oxygen. 5. As these species return to lower-energy states, energy is emitted partly as photons. 6. The CLA detects the resulting extremely weak chemiluminescent emission. 7. The measured emission is used to evaluate the oxidation state or degradation of the material.ケミルミネッセンスアナライザー公式サイトThe important analytical advantage is therefore the detection of oxidation intermediates at an early stage .東北電子産業states that the method can detect oxidation-related changes before conventional observable effects such as significant loss of physical properties or yellowing appear.ケミルミネッセンスアナライザー公式サイトISO later formalized essentially this analytical concept, describing UPE measurement as a method for assessing the very early oxidative degradation state of polymers . ISO

4. Application to Plastics and Recycled

Materials

The technology became particularly important in the analysis of polymeric materials. The company's current technical materials identify applications including:

- accelerated-aging evaluation;
- recycled-material evaluation;
- additive-effect evaluation;
- storage-material acceptance testing;
- weathering evaluation;
- lifetime prediction;
- detection of "silent changes" in materials.ケミルミネッセンスアナライザー公式サイトThe most directly documented industrial development is the METI-supported project:高感度微少酸化計測技術を用いた自動車・情報家電向けエンジニアリングプラスチック材の高効率な再生材利用技術の開発The project focused on engineering plastics for automotive and information-appliance applications and on increasing the efficiency of recycled-material use. Чуco The practical problem is straightforward: recycled polymer can have a different degradation history from virgin material. Repeated thermal processing, extrusion and environmental exposure can increase oxidation, potentially reducing material performance. The weak-light method makes it possible to quantify this degradation and use the information when determining the appropriate

recycled-material mixing ratio. METI explicitly identifies quantitative evaluation of oxidation degradation and determination of recycled-material mixing ratios as project objectives. Чуco

5. Commercialization: From

Measurement Principle to CLA

The technology did not remain a laboratory measurement principle. METI records the development of a specialized weak-light detection apparatus incorporating functions appropriate for plastic and recycled-material evaluation. These included:

- a 350°C-capable high-temperature sample chamber ;
- controlled heating and cooling;
- optimization of analytical conditions;
- dedicated analytical software;
- automatic calculation of oxidation induction time (OIT) ;
- automatic calculation of oxidation onset temperature (OOT) ;
- calculation of activation energy;
- quantitative integration of weak-light measurements with mechanical and physical measurements. Чуco

This demonstrates a transition from: physical phenomenon to measurement technology to industrial analytical instrument . The company continues to market the CLA family as a commercial product and also provides commissioned chemiluminescence measurements. Tei-C This is important from an IP/business perspective because the value is not contained solely in an individual patent claim. The commercial system incorporates detector sensitivity, sample handling, thermal control, measurement methodology and analytical processing. Therefore, the most defensible characterization is: an integrated measurement technology platform protected partly through patents and partly through accumulated technical know-how, subsequently converted into a standardized analytical methodology.

6. Patent / IP Trail

6.1. JP3813140B2

Identification Patent: JP3813140B2 Application: JP2003201271A Title: 「試料の酸化度を測定する方法」 English: Method for measuring the degree of oxidation of a sample Original assignee:東北電子産業株式会社Inventor:木村潤一Priority / filing: 24 July 2003 Publication: JP2005043132A, 17 February 2005 Grant: 23 August 2006. Патенты Google Google Patents currently marks the Japanese patent as expired - fee related , with an anticipated expiration date of 24 July 2023. This status should be treated as database information rather than an independent legal opinion. Патенты Google Relevance The patent concerns measurement of a sample's oxidation degree using emitted light. Its specification describes the use of chemiluminescence for assessing oxidation and deterioration and addresses the difficulty of measuring extremely weak emission. Патенты Google Most importantly for this investigation, the patent record establishes a direct technological relationship with東北電子産業's CLA equipment rather than merely sharing the same corporate applicant. This makes JP3813140B2 a credible early patent-layer component of the technology that eventually became the company's chemiluminescence platform.

6.2. JP2002195951A

Identification Application / publication: JP2002195951A Application: JP2000394936A Title: 「試料の劣化判定方法およびこれに用いる装置」 English: Method for determining deterioration of a sample and apparatus used therefor Original assignee:東北電子産業株式会社Inventors:木村潤一;山田理恵Priority / filing: 26 December 2000 Publication: 10 July 2002. Патенты Google Google Patents currently lists the application as pending . As with the other patent record, this is database status information and should not be treated as a definitive legal opinion without checking the official Japanese patent register. Патенты Google Relevance This patent is particularly important because it concerns determining deterioration through temperature-dependent emission intensity and explicitly involves chemiluminescence-related measurement. The presence of山田理恵as a co-inventor establishes a direct technical connection between the person who later received the JPO/METI recognition and the underlying technology. This does not prove that this particular application caused or determined the 2026 award. It does, however, establish that Yamada's relationship with the relevant technology was not merely managerial or retrospective: she is identified in the patent record as an inventor of an early deterioration-analysis technology.

7. Standardization

7.1. JIS K 7351

The most important standardization milestone is: JIS K 7351 concerning:プラスチックに含まれる過酸化物の微弱発光の高感度測定方法approximately: High-sensitivity measurement method for weak emission from peroxides contained in plastics. METI's official list of projects under the新市場創造型標準化制度identifies: 2016.3 -東北電子産業株式会社

- JIS K 7351 as the standardization project for this measurement method. Минэкономторг Японии

The Tohoku Bureau of METI independently lists the same company and standard and states that JIS K 7351 was publicly issued on 22 January 2018 . Тохоку This is significant because the standardization system was explicitly designed to facilitate standardization of technologies held by individual companies or technologies crossing multiple industry sectors where conventional industry-association processes may not be sufficient. Тохоку In this case, the result was not a standard for the company's particular hardware model. It standardized the measurement method itself . That distinction is central to understanding the strategic importance of the case.

8. International Standardization - ISO

4765:2022

The international successor is: ISO 4765:2022 Title: Chemically-induced ultra-weak photon emission (UPE) - Measurement as an analysis method of degradation of polymeric material ISO identifies the document as an International Standard, first edition, published in April 2022 . ISO Its stated purpose is to provide a method for assessing the very early oxidative degradation state of polymers during outdoor weathering and accelerated indoor weathering tests. It also permits evaluation of the influence of additives. The analytical basis is measurement of chemically induced ultra-weak photon emission. ISO Thus the conceptual relationship is highly consistent: JIS K 7351 → weak emission from peroxides in plastics ISO 4765 → chemically induced UPE as an analytical method for polymer degradation. METI's industrial-standardization award documentation makes the connection even more explicit: it states that東北電子産業developed the Chemiluminescence Analyzer, that it could detect extremely small oxidation-related emissions, that JIS K 7351 was established in 2018 using the New Market Creation-type Standardization System, and that ISO 4765 was established in April 2022. Минэкономторг Японии

9. Business Impact of Standardization

The standardization was not merely academic. METI reports measurable business effects following establishment of the standards. According to the ministry's industrial-standardization award material, after the standards were established:

- commissioned measurement revenue increased from approximately ¥6 million to ¥14 million ;
- participation in the company's research group increased from 50 to 116 participants . METI attributes these changes to increased recognition and credibility of the method

following standardization. Минэкономторг Японии This provides an unusually strong link between: technical IP → standardization → market recognition → measurable commercial effect. The important business mechanism is that a proprietary measurement technology can face a credibility barrier: potential customers may question whether an unusual analytical method is scientifically and commercially established. A JIS/ISO standard changes that situation by giving the methodology an externally recognized measurement framework. In this case, standardization therefore functioned as a market-creation mechanism , not merely as a compliance exercise.

10. Evidence-Based Reconstruction of

the IP Strategy

The available evidence supports the following reconstruction. Stage 1 - Technical invention東北電子産業developed methods for detecting weak emission associated with oxidation and material deterioration. The early patent record includes: JP2002195951A and JP3813140B2 . The former identifies山田理恵as an inventor; the latter identifies東北電子産業as the original assignee. Патенты Google Stage 2 - Productization The technology was incorporated into the company's Chemiluminescence Analyzer product family. The company markets CLA as an ultra-sensitive weak-light detector capable of photon-level measurements. Tohoku Electronic Industrial Co., Ltd. Stage 3 - Industrial specialization The system was adapted to plastic and recycled-material analysis, including high-temperature measurement and software-assisted degradation analysis. METI documents confirm the development of a 350°C-capable chamber and analytical functions for OIT, OOT and activation energy. Чусо Stage 4 - National standardization The measurement methodology became: JIS K 7351 (2018). Тохоку Stage 5 - International standardization The methodology was subsequently formalized internationally as: ISO 4765:2022. ISO Stage 6 - Recognition In 2026, JPO recognized山田理恵personally with the経済産業大臣表彰for contributions to the intellectual-property system. Her own official statement specifically identifies the weak-light material oxidation-degradation evaluation technology and its JIS/ISO standardization as one of the company's major achievements. Японское агентство по патентам

11. Assessment of the Original Research

Question

Question What exactly did東北電子産業develop/protect/commercialize, and why was山田理恵recognized by JPO/METI? Finding東北電子産業developed a chemiluminescence-based ultra-weak photon detection technology for high-sensitivity evaluation of oxidative degradation , with a major application in polymer and plastic materials. The technology was protected in part through patent activity concerning oxidation/degradation measurement and was commercialized as the Chemiluminescence Analyzer (CLA) . The technology was subsequently developed into an industrial analytical platform for plastics and recycled materials. Патенты Google The decisive strategic step was standardization . Rather than keeping the measurement methodology solely as proprietary know-how,東北電子産業used Japan's New Market Creation-type Standardization System to establish JIS K 7351 . The methodology was subsequently internationalized through ISO 4765:2022 , which defines chemically induced ultra-weak photon emission measurement as an analytical method for evaluating polymer degradation. Тохоку The 2026 JPO/METI recognition therefore should not be characterized as an award for a single patent or even solely for the CLA product. The documented basis is broader:山田理恵was recognized personally for her contribution to Japan's intellectual-property system and policy, including practical experience in applying IP and standardization strategy as the head of a regional technology SME. The company's chemiluminescence technology provides a particularly concrete example of that approach because it progressed from technical invention and commercialization to national and international standardization. Японское агентство по патентам

12. Final Analytical Conclusion

The strongest evidence-supported interpretation is:東北電子産業's relevant IP is not a single isolated invention but a technology platform for detecting chemically induced ultra-weak photon emission and using it to quantify early oxidative degradation. The platform was commercialized as the Chemiluminescence Analyzer, applied particularly to polymers and recycled plastics, and supported by an identifiable patent trail. Its strategic importance increased substantially when the measurement methodology was standardized as JIS K 7351 and subsequently ISO 4765:2022. The case is therefore best understood as an example of IP + productization + standardization strategy . The core chain is: Early oxidation/degradation inventions ↓ Patent activity ↓ Chemiluminescence Analyzer (CLA) ↓ Quantitative analysis of polymer/plastic oxidation and recycled materials ↓ JIS K 7351 (2018) ↓ ISO 4765:2022 ↓ Higher recognition and commercial credibility ↓ 2026 JPO/METI recognition of Rie Yamada The evidence does not justify the narrower statement that "JPO awarded Yamada for patent X." It supports a more precise conclusion: the award recognized Yamada's broader contribution to Japan's IP system, while the CLA/UPE technology and its progression into JIS and ISO provide a documented, concrete example of the IP and standardization strategy for which she was recognized.

Sources

Primary / official sources 1. Japan Patent Office -令和8年度「知財功労賞」表彰式Official 2026 award page identifying山田理恵, her position, award category and her own statement concerning the technology. Японское агентство по патентам JPO - 2026 Chizai Koro-sho ceremony 2. Japan Patent Office -令和8年度知的財産権制度関係功労者表彰,山田理恵profile Official PDF detailing the grounds for the award and Yamada's involvement in IP policy and standardization. Японское агентство по патентам 3. METI -新市場創造型標準化制度活用案件一覧Official record identifying東北電子産業's 2016 standardization project and JIS K 7351. Минэкономторг Японии METI - Standardization projects 4. Tohoku Bureau of METI - Industrial Standardization Official regional-government record of東北電子産業and JIS K 7351, including the January 2018 publication date. Тохоку 5. METI - Industrial Standardization Award documentation Official description of the company's CLA, JIS K 7351, ISO 4765 and reported commercial effects after standardization. Минэкономторг Японии 6. METI / Go-Tech -東北電子産業project Official government project record describing the technology, 350°C sample chamber, software, OIT/OOT and recycled-material applications. Чусо 7. ISO - ISO 4765:2022 Official international-standard record and abstract. ISO ISO 4765:2022 Patent / technical sources 8.

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- Chemiluminescence Analyzer Company's technical description of CLA, photon-level detection and JIS/ISO applications. Tohoku Electronic Industrial Co., Ltd.

東北電子産業

- Chemiluminescence Analyzer 11.東北電子産業
- Chemiluminescence mechanism / applications Company's explanation of oxidation, peroxide formation and ultra-weak emission.ケ

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Source-quality note

Highest confidence: JPO award classification, JPO's stated grounds, METI standardization records, ISO 4765 existence/title/purpose, and METI's description of the company's CLA and standardization results. These are primary institutional sources. High confidence: the identification of JP3813140B2 and JP2002195951A and their relationship to東北電子産業, because the patent records explicitly identify the applicant/assignee and technical subject matter. Патенты Google Important limitation: the available evidence supports the conclusion that these patents are relevant components of the technology's IP history , but it does not establish that either patent alone legally covers the entirety of the current CLA product. The modern commercial system also incorporates hardware, measurement procedures, software and know-how developed over time. This distinction should be retained in any portfolio presentation.